Lecture 19

Electrical Properties

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Week 12 lectures

Electrical properties

- Conduction
- Semi-conduction
- Dielectric behaviour
- Ferroelectric behaviour

Thermal properties

- Heat capacity
- Heat conduction
- Thermo-mechanical behaviour

Background information:

For this week's lectures read

Callister and Rethwisch 9e   Chapter 19 and Chapter 20

Aims

&

Learning Outcomes

DISCUSS THE PROCESS OF CONDUCTION OF ELECTRICITY IN MATERIALS

DESCRIBE ELECTRICAL CONDUCTIVITY IN SEMICONDUCTING MATERIALS.

DESCRIBE THE FOUR POSSIBLE ELECTRON BAND STRUCTURES FOR SOLID MATERIALS

EXPLAIN THE PHENOMENON OF SEMICONDUCTIVITY, INTRINSIC AND EXTRINSIC SEMICONDUCTOR

Electrical conduction

- **Ohm's law**

$$V = IR \qquad (1)$$

where:

V = **electrical potential, volts, V** (J/C)
or applied voltage

I = **electrical current, amperes, A** (C/s)
the electrical charge passing / time

R = **resistance, ohms (Ω)**
the resistance of the material to electrical current

Georg Ohm
German scientist
1789-1854

Electrical conduction

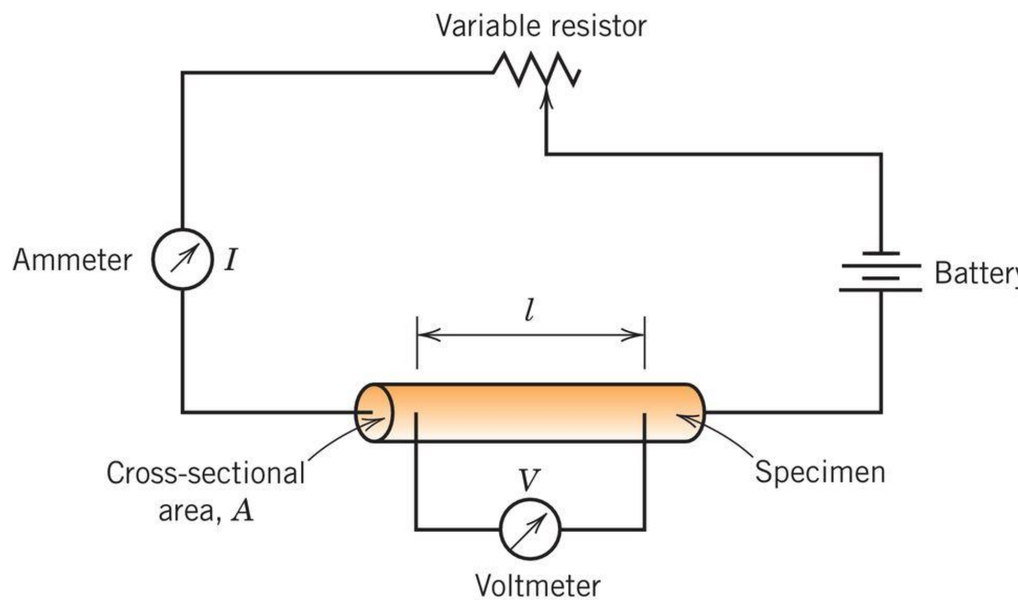
- **Resistivity** is independent of specimen size and geometry

A = Area, not amperes

$$\rho = \frac{RA}{l} \qquad (2)$$

or

$$\rho = \frac{VA}{Il} \qquad (3)$$



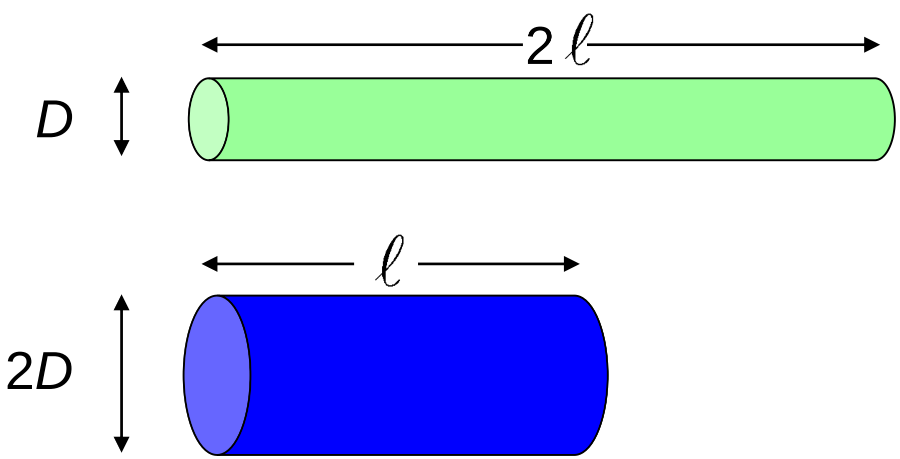
Resistivity has **units** of ohm x m (Ω.m)

Fig 19.1 Callister & Rethwisch 9e

Conventionally, +ve current flows from positive potential to negative (in reality, -ve electrons flow from negative to positive)

Electrical conduction

- Resistance depends on specimen geometry and size, using (2)



$$R_1 = \frac{2\rho\ell}{\pi\left(\frac{D}{2}\right)^2} = \frac{8\rho\ell}{\pi D^2}$$

$$R_2 = \frac{\rho\ell}{\pi\left(\frac{2D}{2}\right)^2} = \frac{\rho\ell}{\pi D^2} = \frac{R_1}{8}$$

Resistance of the first conductor is 8x that of the second

Electrical conductivity

- **Conductivity**,

the ease with which electrical current flows in a material
is the inverse of resistivity

$$\sigma = \frac{1}{\rho} \qquad (4)$$

units of inverse ohm x m $(\Omega.m)^{-1}$

Ohm's law can also be expressed in terms of **current density**, J
(substitute (2) and (4) into (5) to recover Ohm's law (1))

$$J = \sigma \mathcal{E} \qquad (5)$$

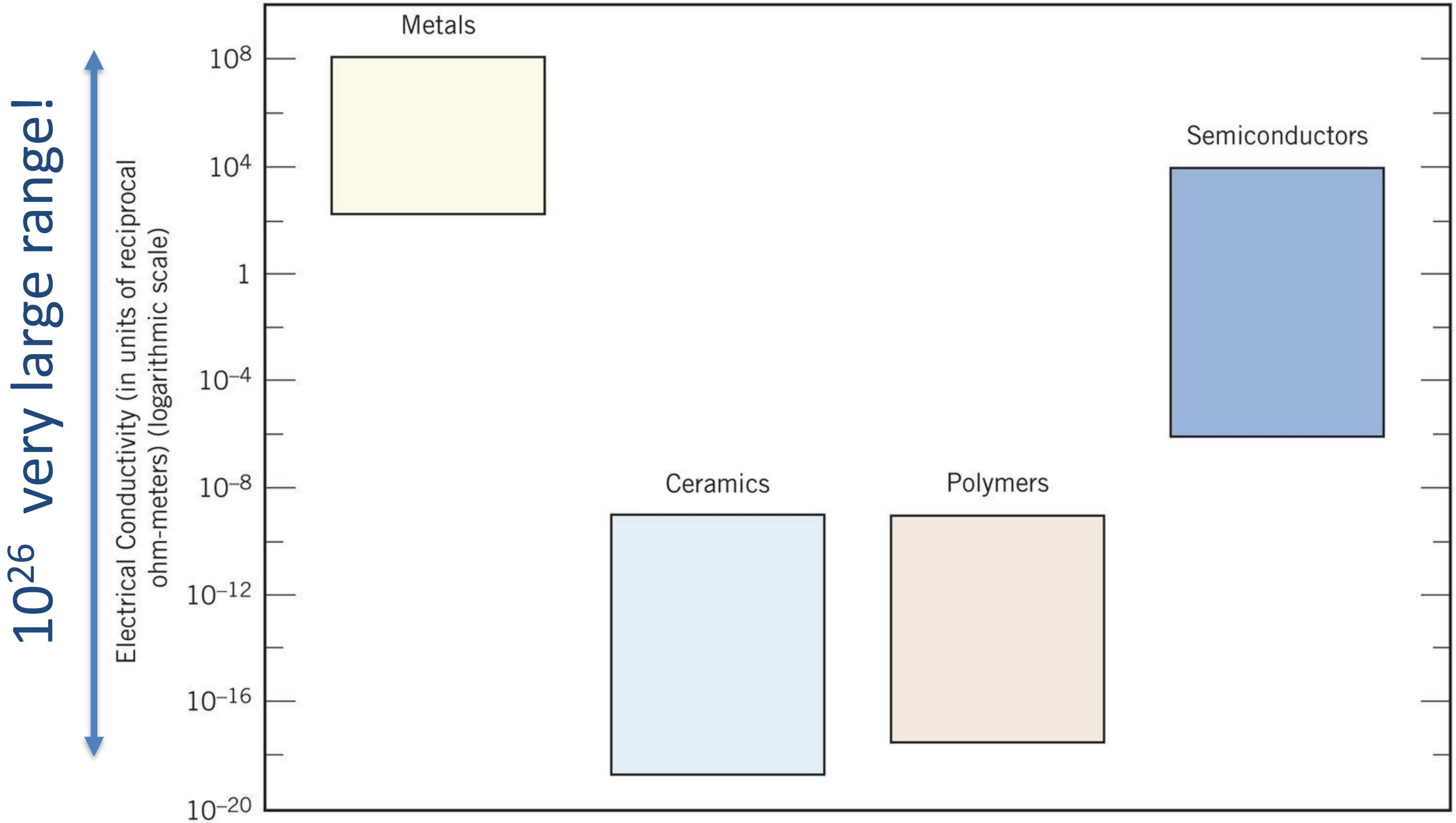
units of amperes / m^2 (A/m^2)

where $J = I / A$
and $\mathcal{E}$ is the **electric field strength** $\mathcal{E} = \frac{V}{l}$
$\mathcal{E}$ has units of V/m

Electrical properties

- conductivity of materials

Fig 1.8 Callister & Rethwisch 9e



Bar chart of room temperature electrical conductivity ranges for metals, ceramics, polymers, and semiconducting materials.

Conductivity: Comparison

- Room temperature values $(\text{Ohm-m})^{-1} = (\Omega \text{ - m})^{-1}$

METALS — conductors

Silver	6.8×10^{7}
Copper	6.0×10^{7}
Iron	1.0×10^{7}

CERAMICS

Soda-lime glass	10^{-10}-10^{-11}
Concrete	10^{-9}
Aluminum oxide	$<10^{-13}$

SEMICONDUCTORS

Silicon	4×10^{-4}
Germanium	2×10^{0}
GaAs	10^{-6}

semiconductors

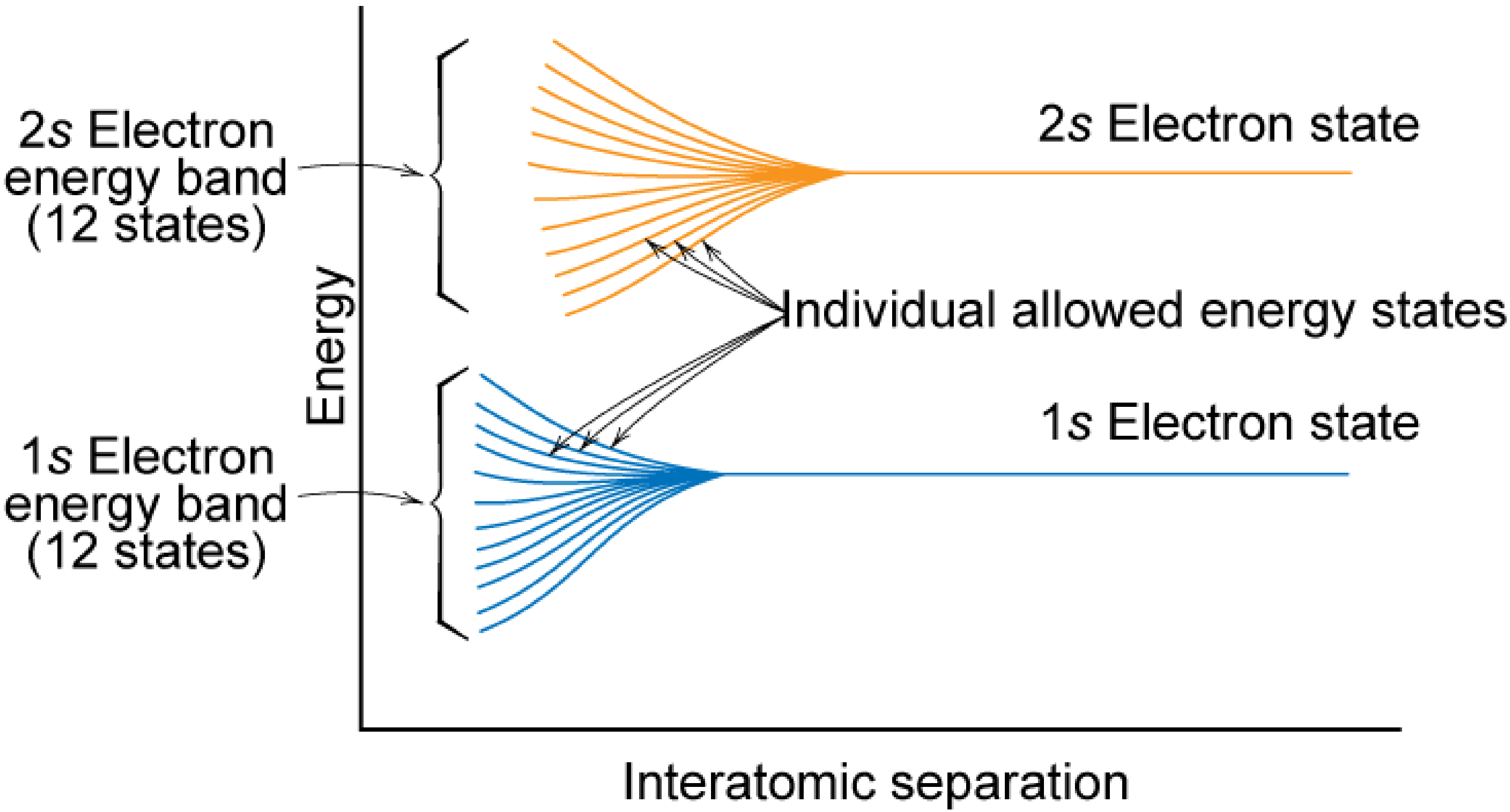
POLYMERS

Polystyrene	$<10^{-14}$
Polyethylene	10^{-15}-10^{-17}

insulators

Selected values from Tables 18.1, 18.3, and 18.4, *Callister & Rethwisch 8e.*

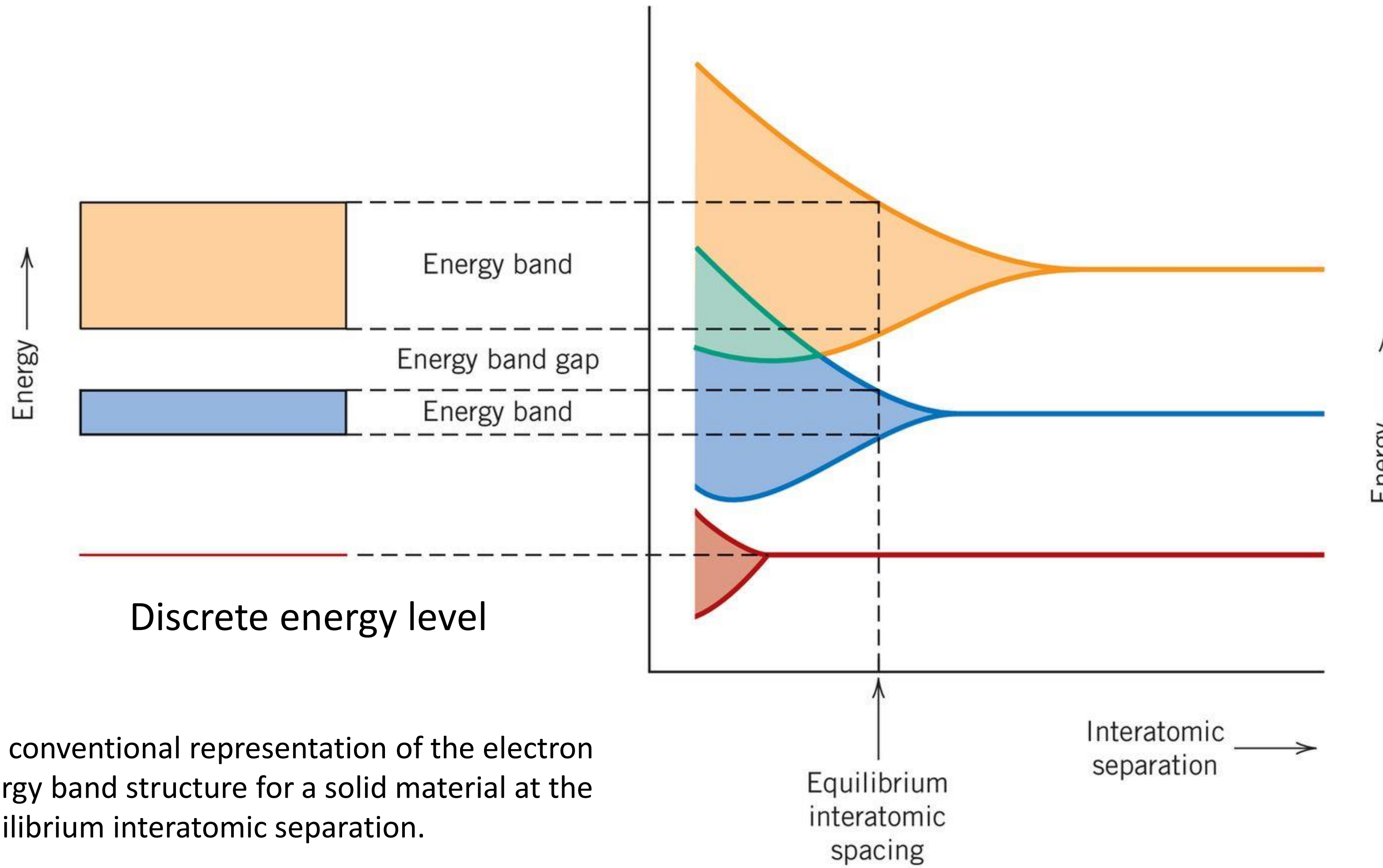
Electron Energy Band Structures



Adapted from Fig. 18.2, *Callister & Rethwisch 8e.*

Energy band structures in solids

- Electron band structure



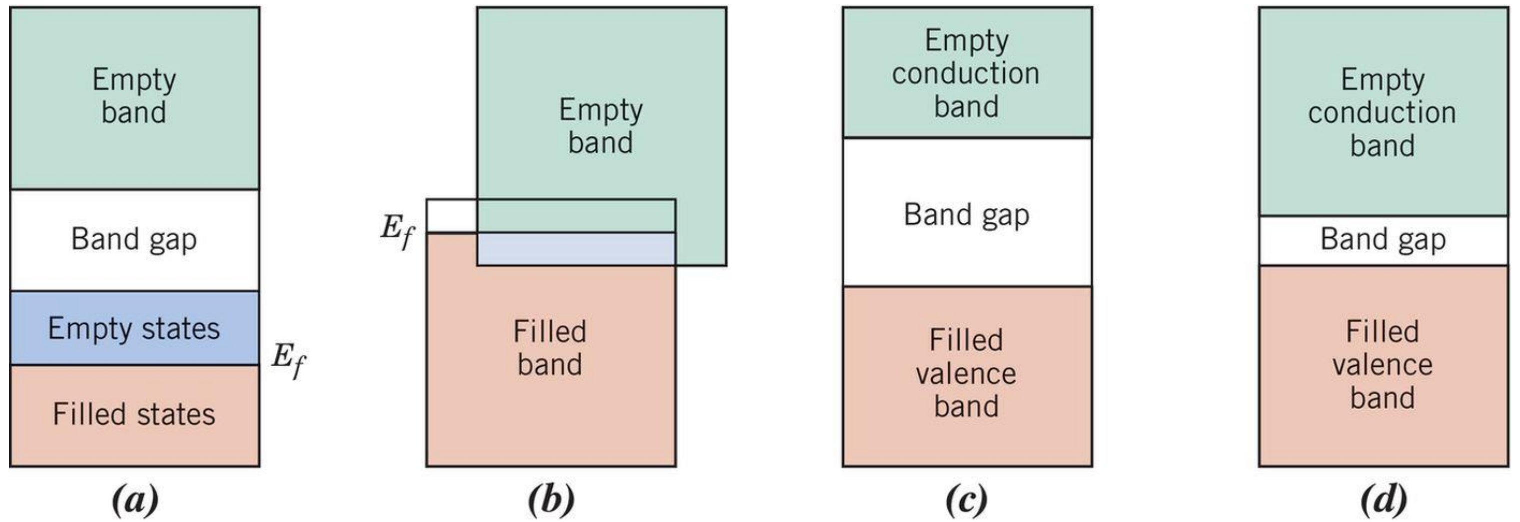
a) The conventional representation of the electron energy band structure for a solid material at the equilibrium interatomic separation.

b) Electron energy versus interatomic separation for an aggregate of atoms, illustrating how the energy band structure at the equilibrium separation in (a) is generated.

Fig 19.3 Callister & Rethwisch 9e

Energy band structures in solids

Fig 19.4 Callister & Rethwisch 9e



Metal e.g. Cu Metal e.g. Mg Insulator Semi-conductor

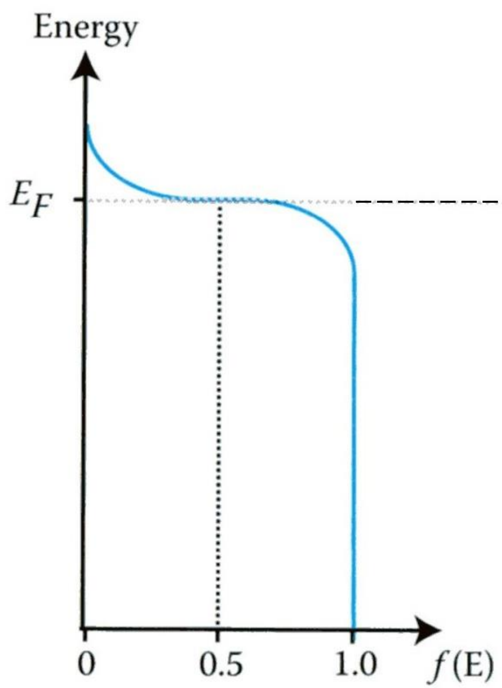
- **Metals** have adjacent empty states = conduct electrons easily
- **Insulators** have a wide 'band gap' > 2eV
- **Semi-conductors** have a narrow band gap < 2eV

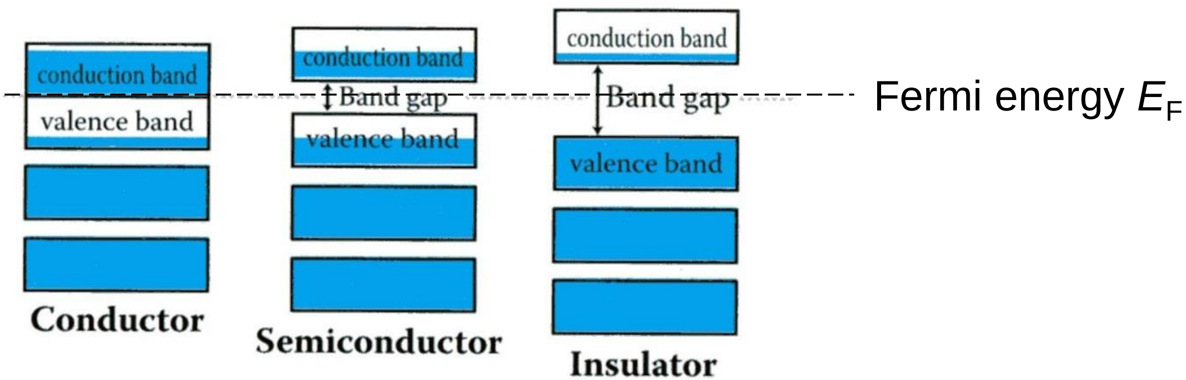
eV, electron volts is a unit of energy,

J = VC **= charge on an electron (1.6×10^{-19} C) x voltage**

Fermi energy

- the kinetic energy of the highest occupied state at zero Kelvin





Fermi function
$$f(\varepsilon) = \frac{1}{e^{(\varepsilon - \varepsilon_F)/k_B T} + 1}$$

The Fermi function shows that at all temperatures above zero Kelvin thermal energy means that there is a probability of electrons jumping to the conduction band.

Electron mobility

- electrons do not move easily through solids in response to an electrical potential (applied voltage) they are scattered by interactions with atoms in the solid

The **drift velocity** is

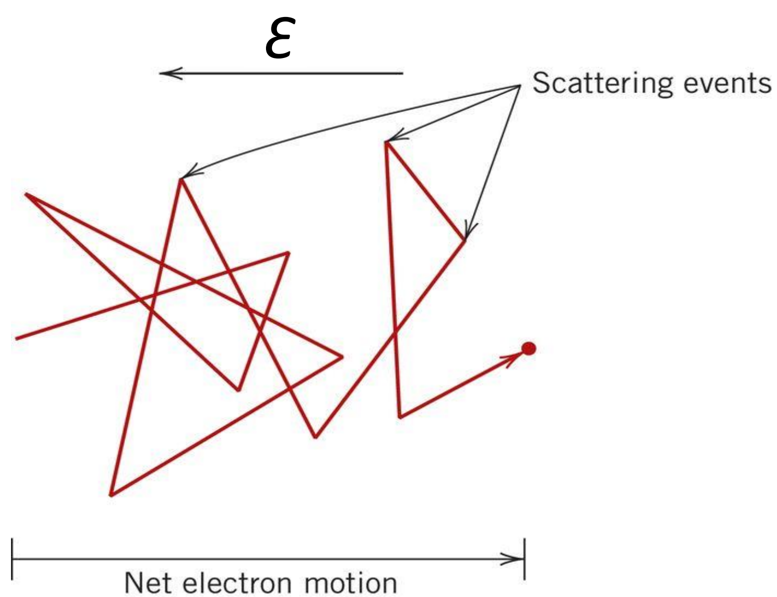
$$v_d = \mu_e \mathcal{E}$$

where μ_e is the **electron mobility** with **units** of $m^2/V.s$

Conductivity can be expressed as

$$\sigma = n\,|e|\mu_e$$

where *n* is the number of conducting electrons / volume and $|e|$ is the absolute charge on an electron



Path of electrons deflected by scattering events. Fig 19.7 Callister, 9e.

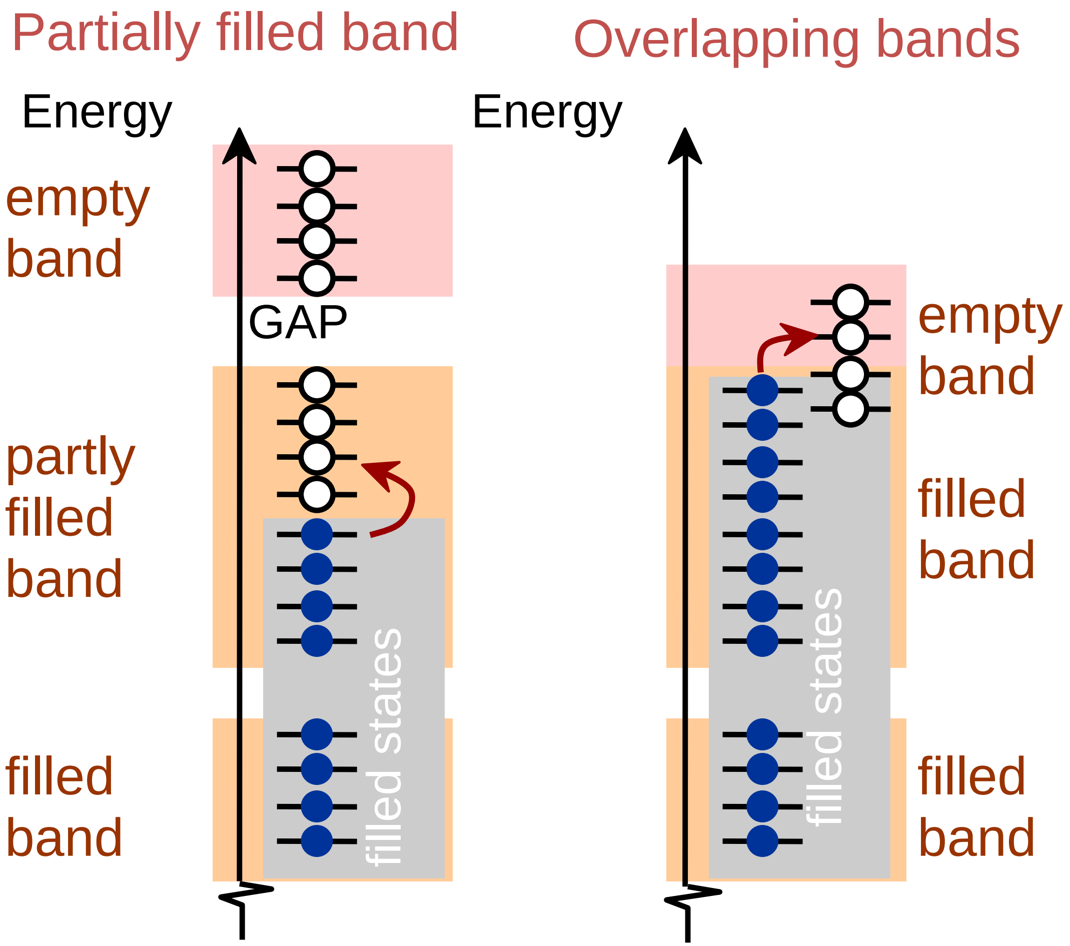
Conduction & Electron Transport

- Metals (Conductors):

-- for metals empty energy states are adjacent to filled states.

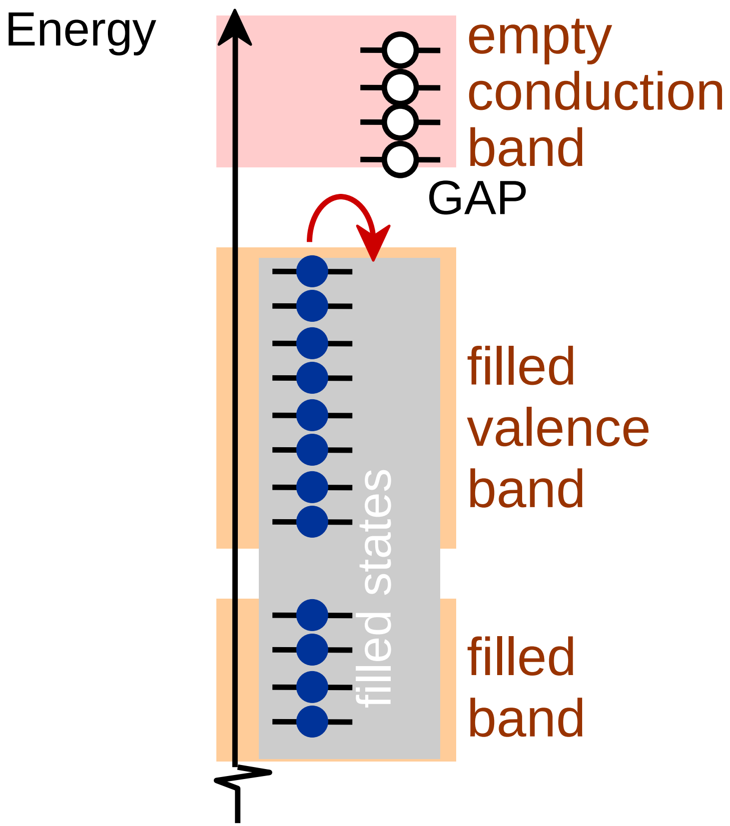
-- thermal energy excites electrons into empty higher energy states.

-- two types of band structures for metals
- partially filled band
- empty band that overlaps filled band

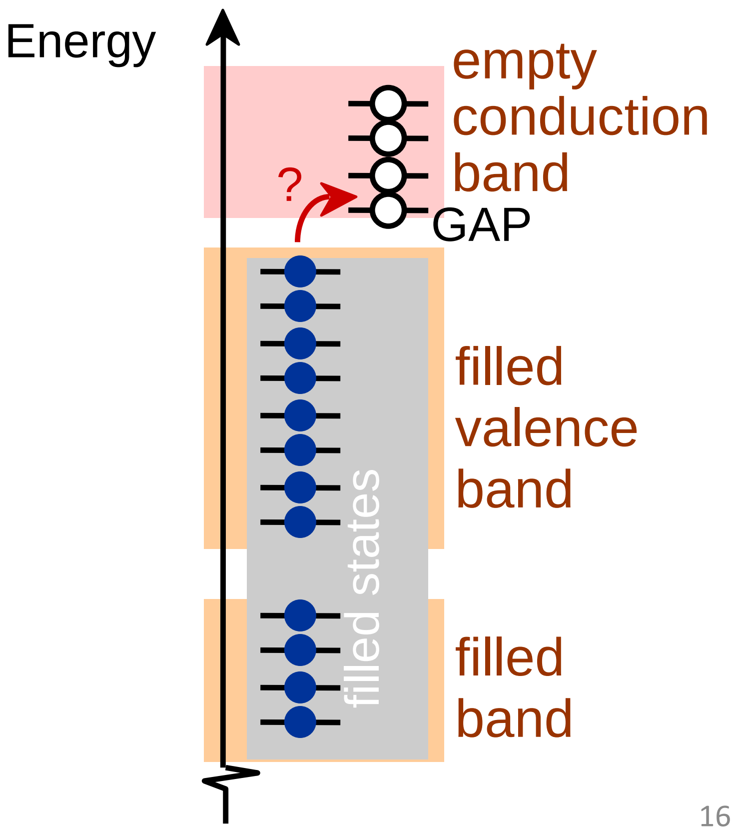


Energy Band Structures: Insulators & Semiconductors

- Insulators:
  -- wide band gap (> 2 eV)
  -- few electrons excited across band gap



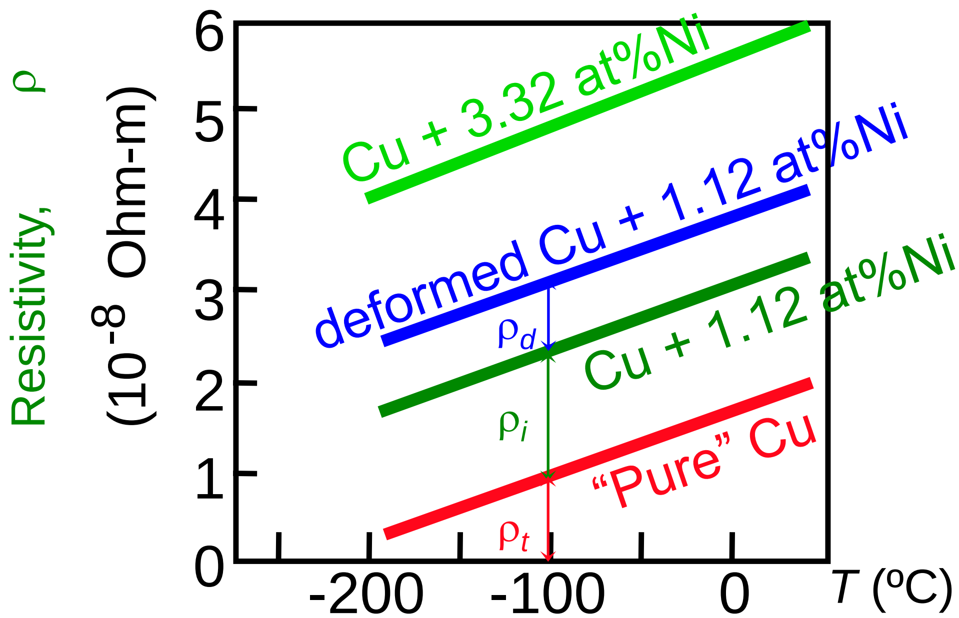
- Semiconductors:
  -- narrow band gap (< 2 eV)
  -- more electrons excited across band gap



Metals: Influence of Temperature and Impurities on Resistivity

- Presence of imperfections increases resistivity
  - -- grain boundaries
  - -- dislocations
  - -- impurity atoms
  - -- vacancies

These act to scatter electrons so that they take a less direct path.



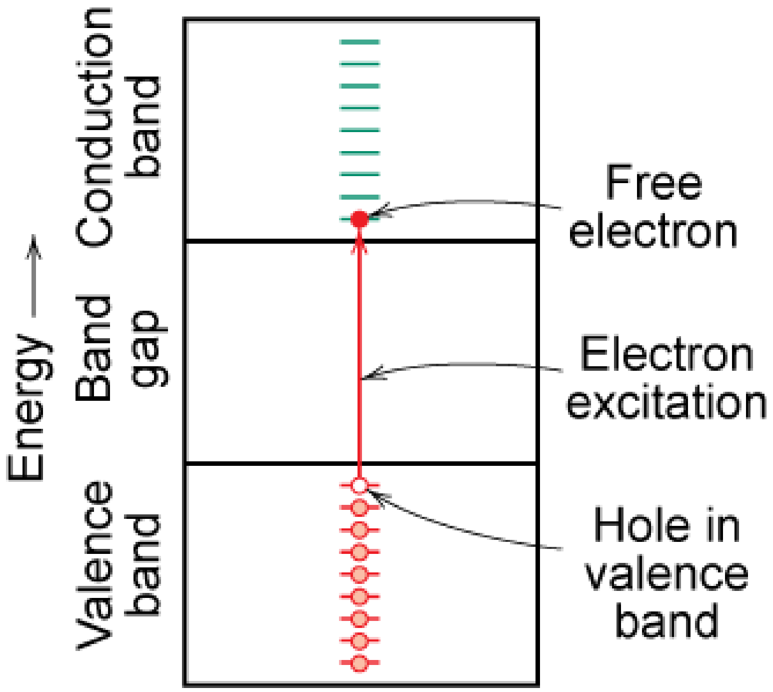
Adapted from Fig. 18.8, *Callister & Rethwisch 8e.* (Fig. 18.8 adapted from J.O. Linde, *Ann. Physik* **5**, p. 219 (1932); and C.A. Wert and R.M. Thomson, *Physics of Solids*, 2nd ed., McGraw-Hill Book Company, New York, 1970.)

- Resistivity increases with:
  - -- temperature
  - -- wt% impurity
  - -- %*CW*

$$\rho = \rho_{thermal} + \rho_{impurity} + \rho_{deformation}$$

Charge Carriers in Insulators and Semiconductors

Adapted from Fig. 19.6(b), *Callister & Rethwisch 9e.*



Two types of electronic charge carriers:

Free Electron

- negative charge
- in conduction band

Hole

- positive charge
- vacant electron state in the valence band

Move at different speeds - drift velocities

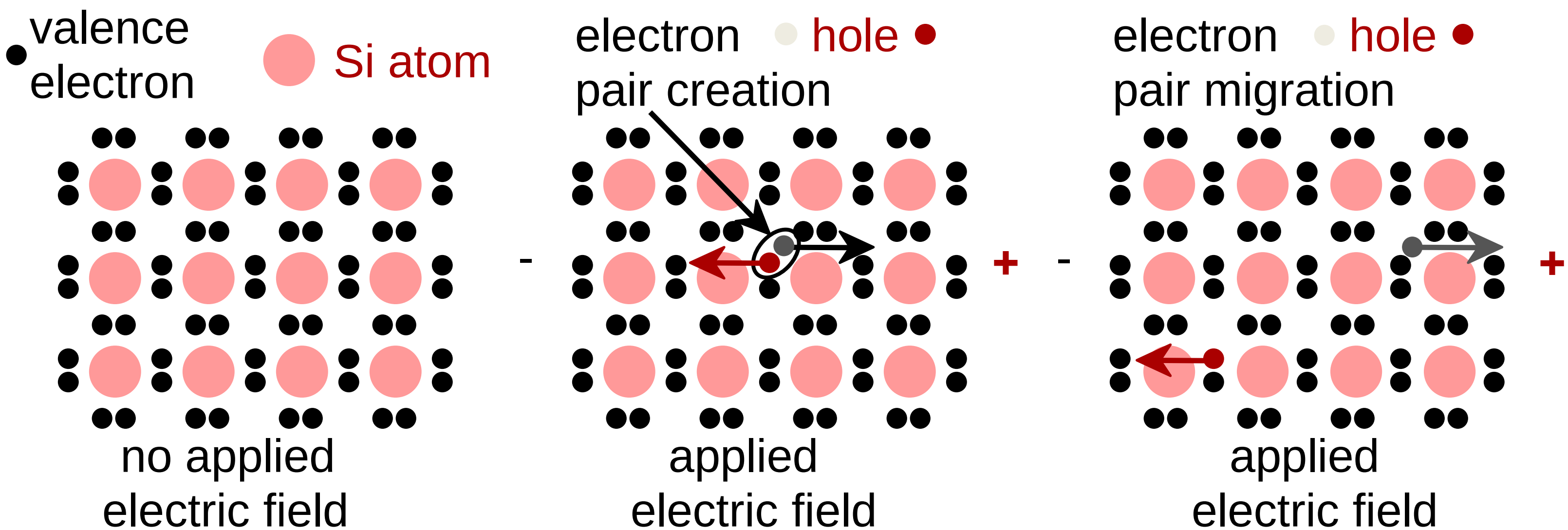
Intrinsic Semiconductors

- Pure material semiconductors: e.g., silicon & germanium
  - Group IVA materials

- Compound semiconductors
  - III-V compounds
    - Ex: GaAs & InSb
  - II-VI compounds
    - Ex: CdS & ZnTe
  - The wider the electronegativity difference between the elements the wider the energy gap.

Intrinsic Semiconduction in Terms of Electron and Hole Migration

- Concept of electrons and holes:



Adapted from Fig. 18.11, *Callister & Rethwisch 8e.*

- Electrical Conductivity given by:

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

holes/m³

hole mobility

electrons/m³

electron mobility

Number of Charge Carriers

Intrinsic Conductivity

$$\boxed{\sigma = n|e|\mu_e + p|e|\mu_h}$$

n = number of electrons
p = number of holes
n_i = intrinsic carrier concentration
μ_e & μ_h = electron and hole mobility

- for intrinsic semiconductor $n = p = n_i$

$$\therefore \qquad \sigma = n_i|e|(\mu_e + \mu_h)$$

Example: GaAs

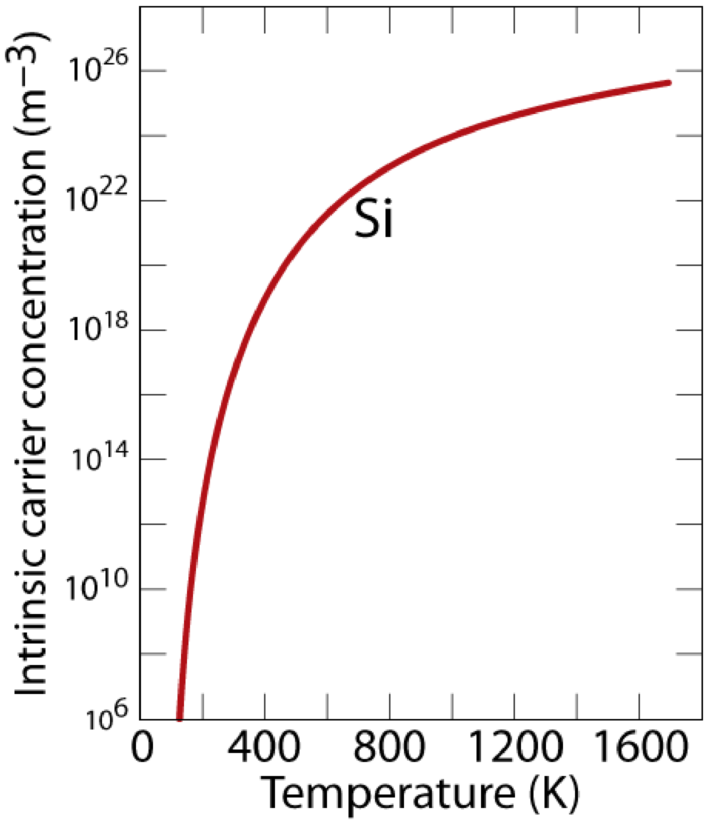
$$n_i = \frac{\sigma}{|e|(\mu_e+\mu_p)} = \frac{3\times10^{-7}\ (\Omega m)^{-1}}{(1.6\times10^{-19}\mathrm{C})(0.80+0.04\ \mathrm{m^2/Vs})}$$

For GaAs $n_i = 2.2 \times 10^{12}\ \mathrm{m^{-3}}$

For Si $n_i = 1.1 \times 10^{16}\ \mathrm{m^{-3}}$

Intrinsic Semiconductors: Conductivity vs *T*

- Data for Pure Silicon:
  - -- σ increases with *T*
  - -- opposite to metals



Adapted from Fig. 18.16, *Callister & Rethwisch 8e.*

$$\sigma = n_i |e| (\mu_e + \mu_h)$$

$$n_i \propto e^{-E_{gap}/kT}$$

material	band gap (eV)
Si	1.11
Ge	0.67
GaP	2.25
CdS	2.40

Selected values from Table 18.3, *Callister & Rethwisch 8e.*

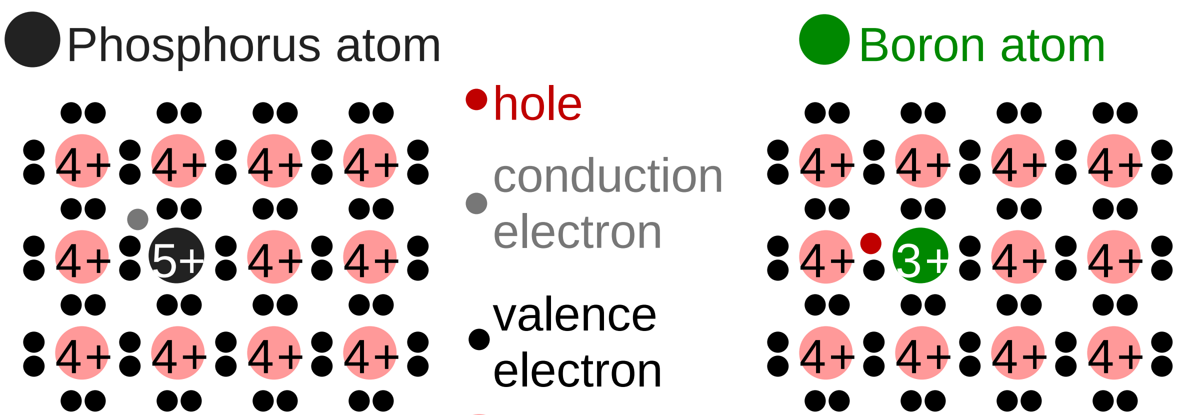
Intrinsic vs Extrinsic Conduction

- Intrinsic:
  -- case for pure Si
  -- # electrons = # holes ($n = p$)

- Extrinsic:
  -- electrical behavior is determined by presence of impurities that introduce excess electrons or holes
  -- $n \neq p$

- *n*-type Extrinsic: ($n >> p$)
- *p*-type Extrinsic: ($p >> n$)

$$\sigma \approx n|e|\mu_e$$



no applied electric field

no applied electric field

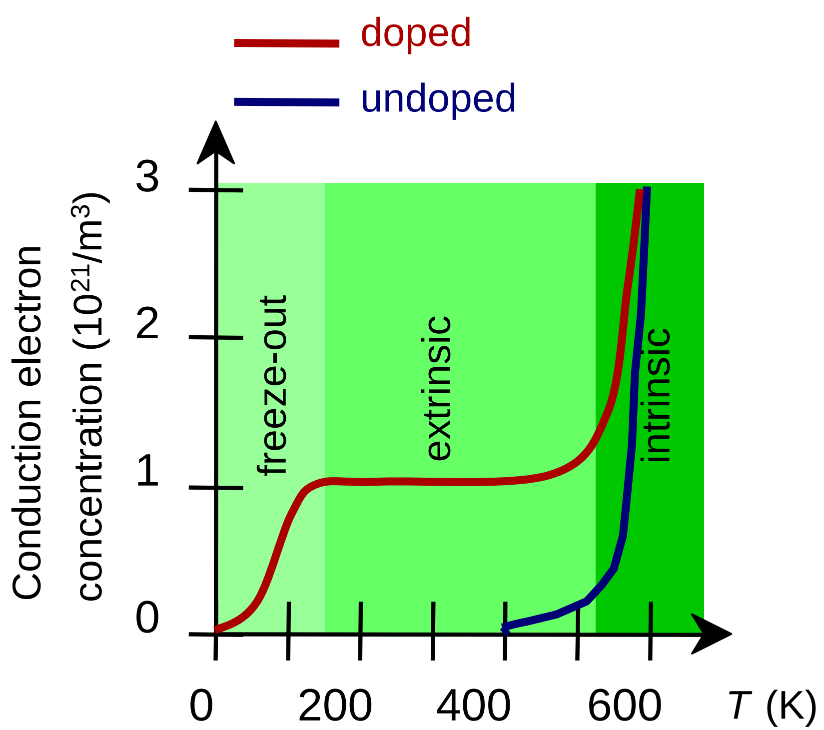
$$\sigma \approx p|e|\mu_h$$

Adapted from Figs. 18.12(a) & 18.14(a), *Callister & Rethwisch 8e.*

Extrinsic Semiconductors: Conductivity vs. Temperature

- Data for Doped Silicon:
  - -- σ increases doping
  - -- reason: imperfection sites lower the activation energy to produce mobile electrons.

- Comparison: intrinsic vs extrinsic conduction...
  - -- extrinsic doping level: $10^{21}/m^3$ of a *n*-type donor impurity (such as P).
  - -- for *T* < 100 K: "freeze-out“, thermal energy insufficient to excite electrons.
  - -- for 150 K < *T* < 450 K: "extrinsic"
  - -- for *T* >> 450 K: "intrinsic"

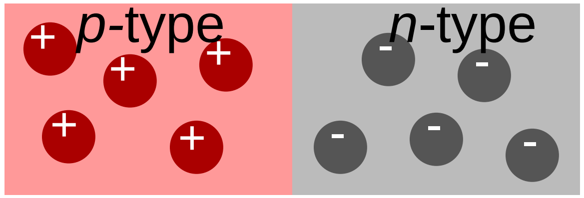


Adapted from Fig. 18.17, *Callister & Rethwisch 8e.* (Fig. 18.17 from S.M. Sze, *Semiconductor Devices, Physics, and Technology*, Bell Telephone Laboratories, Inc., 1985.)

p-n Rectifying Junction

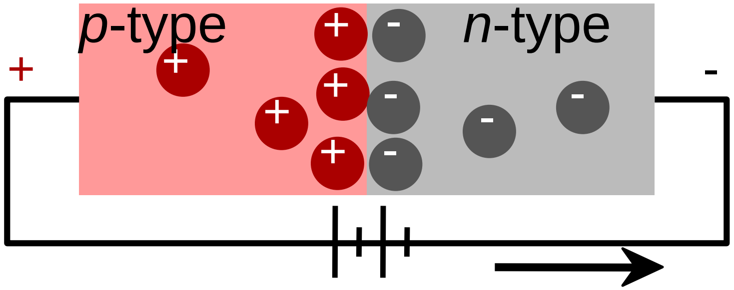
- Allows flow of electrons in one direction only (e.g., useful to convert alternating current to direct current).
- Processing: diffuse P into one side of a B-doped crystal.

-- No applied potential: no net current flow.

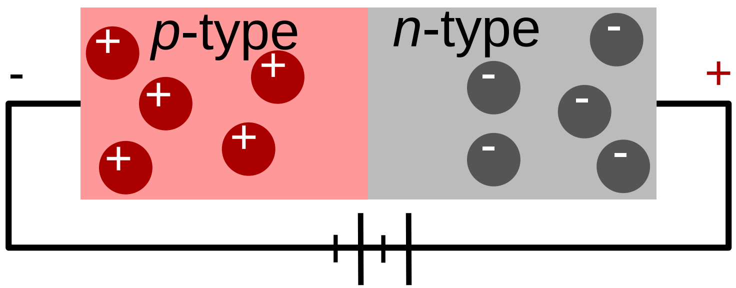


Adapted from Fig. 18.21 Callister & *Rethwisch 8e.*

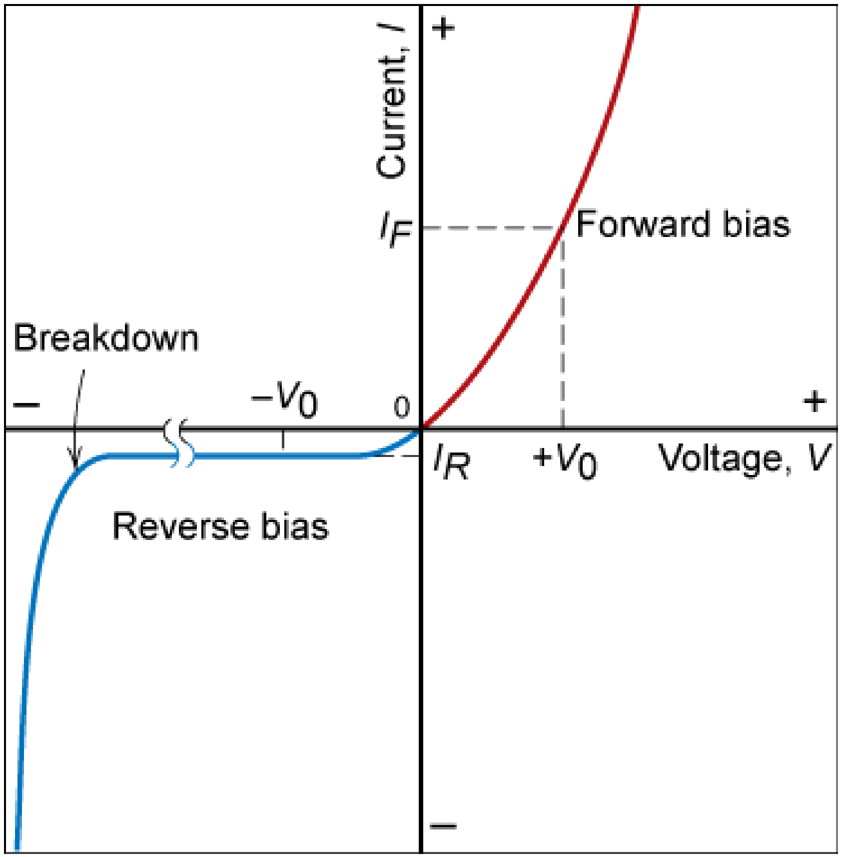
-- Forward bias: carriers flow through *p*-type and *n*-type regions; holes and electrons recombine at *p-n* junction; current flows.



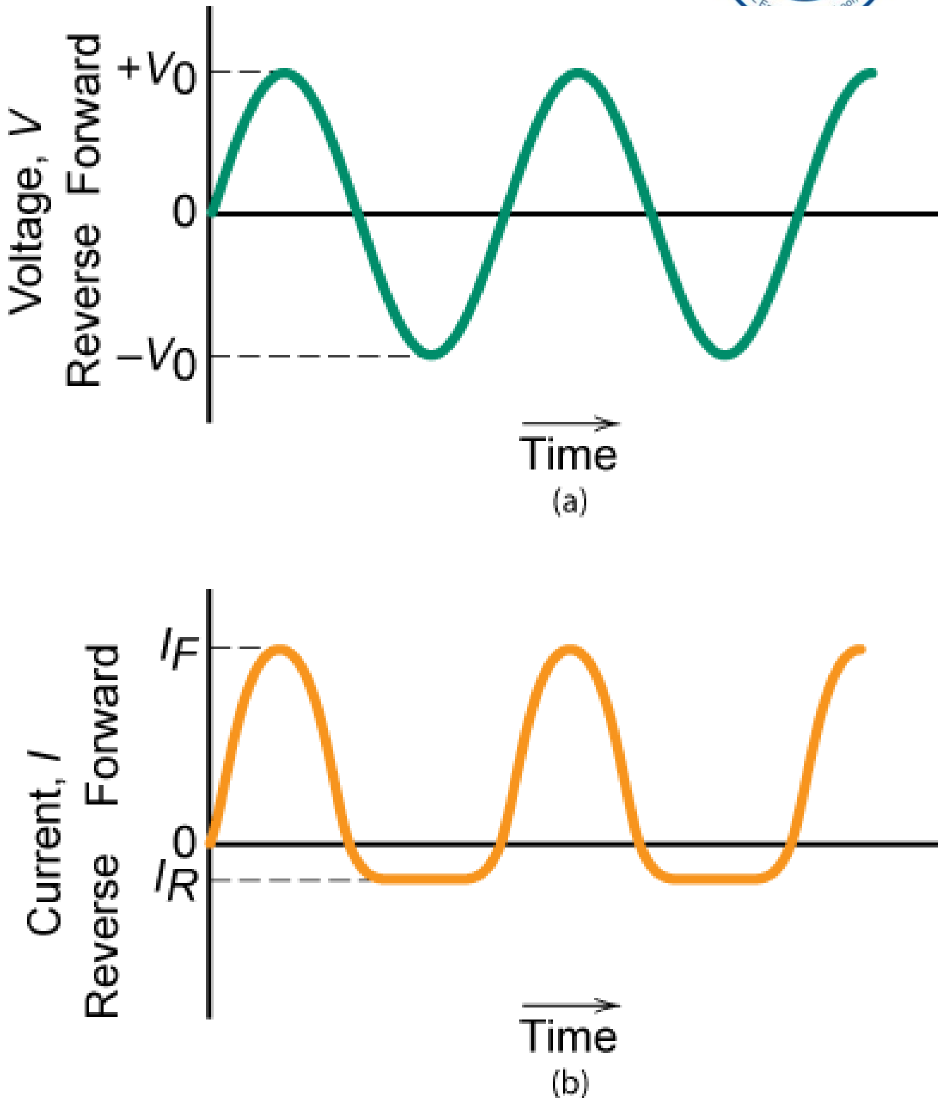
-- Reverse bias: carriers flow away from *p-n* junction; junction region depleted of carriers; little current flow.



Properties of Rectifying Junction

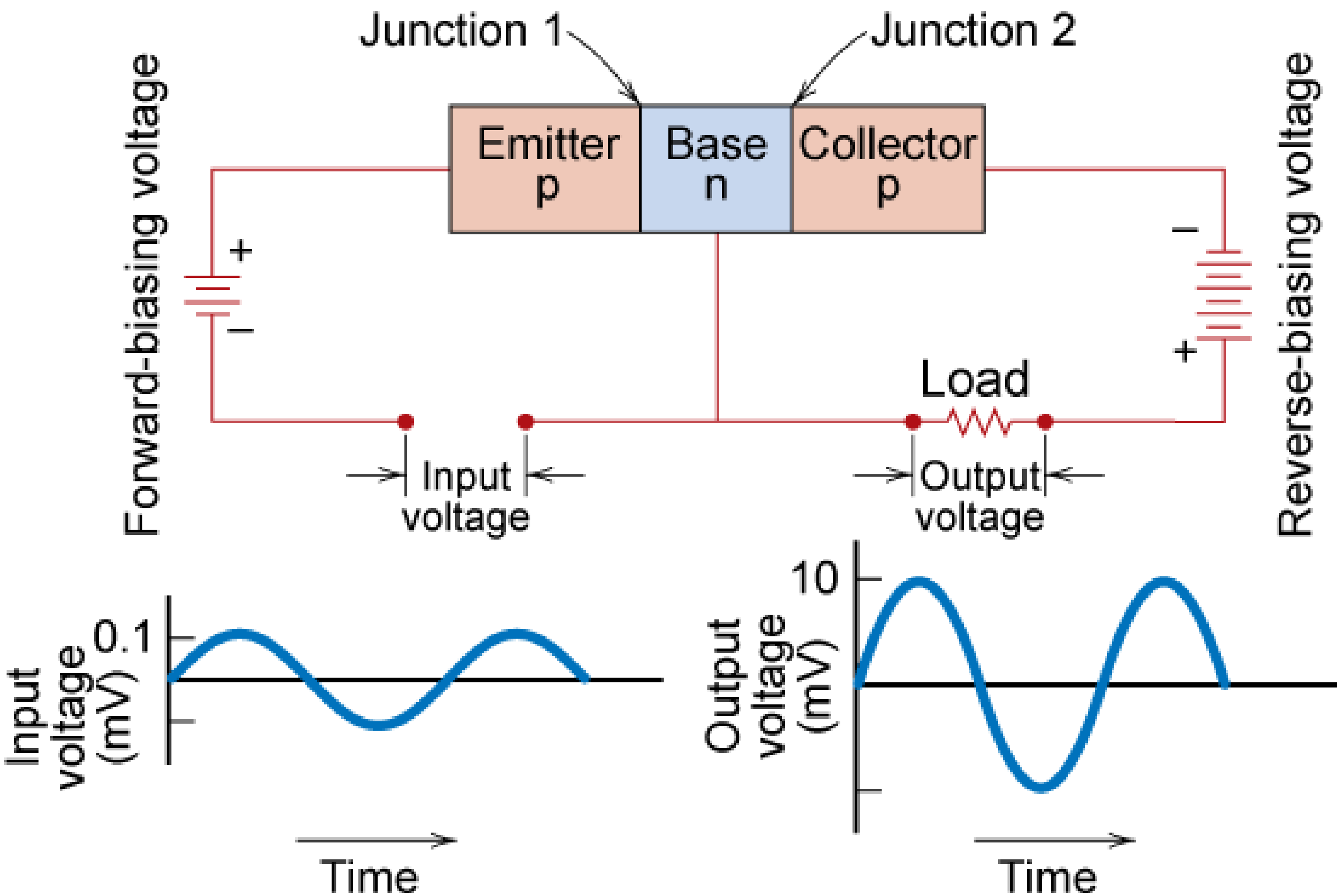


Current-voltage characteristics of a p-n junction for forward and reverse biases. Fig. 19.22, *Callister & Rethwisch 9e.*



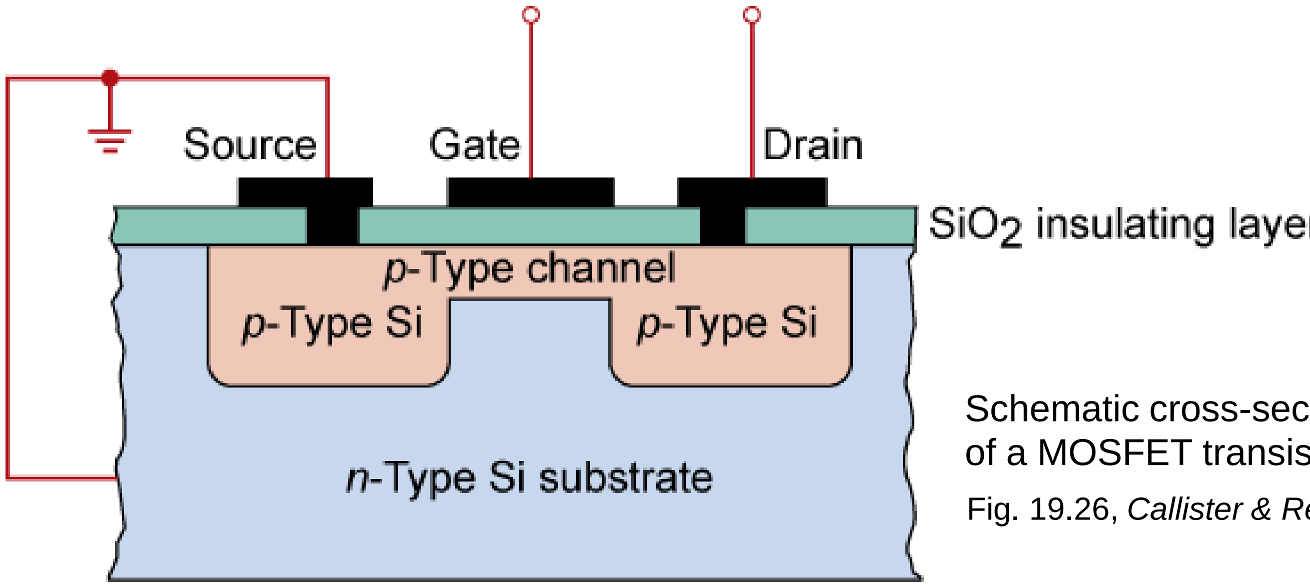
Top: Voltage vs T for the input to a p-n rectifying junction.
Bottom: current vs time showing rectification of voltage.
Fig. 19.23, *Callister & Rethwisch 9e.*

Junction Transistor



Schematic diagram of a *p–n–p* junction transistor and its associated circuitry, including input and output voltage–time characteristics showing voltage amplification. Fig. 19.24, *Callister & Rethwisch 9e.*

MOSFET Transistor
Integrated Circuit Device



Schematic cross-sectional view of a MOSFET transistor

Fig. 19.26, *Callister & Rethwisch 8e.*

- MOSFET (metal oxide semiconductor field effect transistor)

- Application = switch, logic, memory

Summary

- Electrical *conductivity* and *resistivity* are:

  -- material parameters

  -- geometry independent
- Conductors, semiconductors, and insulators...

  -- differ in range of conductivity values

  -- differ in availability of electron excitation states
- For metals, *resistivity* is increased by

  -- increasing temperature

  -- addition of imperfections

  -- plastic deformation
- For pure semiconductors, *conductivity* is increased by

  -- increasing temperature

  -- doping [e.g., adding B to Si (*p*-type) or P to Si (*n*-type)]

Additional learning material

- Review content in DoItPoms website:

https://www.doitpoms.ac.uk/tlplib/thermal_electrical/intro_equations.php

https://www.doitpoms.ac.uk/tlplib/semiconductors/index.php

- Review concepts from Physics on electron mobility and drift velocity

- Watch videos of:
  - Ferroelectricity
  - Piezoelectric materials

in QMPlus